

Taeyoung Sohn

Name: Taeyoung Sohn (손태영)

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I am a machine learning engineer who specializes in bridging the gap between state-of-the-art research and scalable, production-ready products. My work spans from recommendation systems and generative AI to computer vision, combining strong research experience with hands-on engineering expertise.

Publications

WEDGE: Web-Image Assisted Domain Generalization for Semantic Segmentation

Namyup Kim, **Taeyoung Son**, Jaehyun Pakk, Cuiling Lan, Wenjun Zeng, Suha Kwak

IEEE International Conference on Robotics and Automation (ICRA), 2023

arxiv: <https://arxiv.org/abs/2109.14196>

FIFO: Learning Fog-Invariant Features for Foggy Scene Segmentation

Sohyun Lee, **Taeyoung Son**, Suha Kwak

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2022 (Oral, Best Paper Finalist)

arxiv: <https://arxiv.org/abs/2204.01587>

URIE: Universal Image Enhancement for Visual Recognition in the Wild

Taeyoung Son, Juwon Kang, Namyup Kim, Sunghyun Cho, Suha Kwak

European Conference on Computer Vision (ECCV), 2020

arxiv: <https://arxiv.org/abs/2007.08979>

Patent

Method and device for learning fog-invariant feature

- Sohyun Lee, Suha Kwak, **Taeyoung Son**
- US Patent number: [US20230419654A1](#)

Skills

Generative AI(Stable Diffusion, FLUX), Recommendation System, Large Language Model, Machine Learning, Apache Spark, Apache Airflow, AWS, Python, Pytorch, SQL

Education

Mar. 2020 – Mar. 2022

M.S., POSTECH (Pohang University of Science and Technology)

- Department of Computer Science and Engineering (CSE)
- Computer Vision Laboratory
- Thesis: *Universal Image Enhancement for Robust Visual Recognition*
- Advisor: Prof. Suha Kwak

Mar. 2015 – Feb. 2020

B.S., POSTECH (Pohang University of Science and Technology)

- Department of Computer Science and Engineering (CSE)
 - Thesis: *Visual recognition under extreme condition*
 - Advisor: Prof. Suha Kwak
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Careers

Machine Learning Engineer at RIDI

Oct. 2023 – Present

Developed a recommendation system for the 'For You' section of the Manta

Apr. 2025 - Present

- Role: Architected the end-to-end MLOps pipeline for personalized recommendation modeling and serving.
- Impacts:
 - Increased the recommendation tab's Click-Through Rate (CTR) by **13 percentage points**.
 - Improved view-to-read, view-to-pay rate by significant amount.

Developed the 'Trending Now' recommendation feature for the RIDIBOOKS

Jan. 2025 - Apr. 2025

- Role: Built an automated data pipeline using Spark to aggregate and serve real-time trending content.
- Impact: Boosted the search page's conversion rate by **1.79%p** and paid conversion rate by **1.00%p**.

Developed a chatbot to enhance internal knowledge accessibility

Oct.2024 - Feb. 2025

- **Role:** Created an internal AI assistant with real-time translation and RAG-based Q&A capabilities.
- **Impact:** Reduced time spent on repetitive internal inquiries and improved cross-lingual team communication.

Developed an image generation algorithm for the Prodifi

Oct.2023 - Oct. 2024

- **Role:** Implemented image inpaint pipelines using ControlNet and LoRA to support content creation automation.
- **Impact:** Significantly improved the efficiency of the content team's image creation and localization workflow.

AI Research Scientist at NALBI

Apr. 2022 – Oct. 2023

Developed a real-time 3D human body reconstruction algorithm for the A.PLA

- **Role:** Implemented **post-training quantization** to optimize a 3D human reconstruction model for real-time, on-device performance.
- **Impact:** Achieved over **30 FPS** in inference performance across various mobile devices.

Research Scientist Internship at HyperConnect

Jun. 2018 – Sep. 2018, Jun. 2019 – Aug. 2019

Trained a classification model to filter inappropriate profile pictures in the Azar

Jun. 2019 – Aug. 2019

- **Role:** Developed an image classification model to automatically filter content from a large-scale, imbalanced dataset.
- **Impact:** Achieved **96.51% accuracy** on a multi-million image dataset.

Developed a model to generate Look-Up Tables (LUTs) for on-device camera beautification filters

Jun. 2018 – Sep. 2018

- **Role:** Developed a model to generate lightweight Look-Up Tables (LUTs) for on-device image enhancement.
- **Impact:** Developed a neural network to generate LUTs and validated their effectiveness through internal A/B testing.

Personal Projects

A collection of side projects based on personal interests.

Rotary Position Embedding in Triton

- **Core:** Implemented a low-level, optimized GPU kernel for Rotary Position Embedding (RoPE) using the Triton language.
- **Outcome:** Benchmarked and validated performance improvements against the standard PyTorch implementation, deepening understanding of kernel optimization.
- **Github:** https://github.com/taeyoungson/triton_rope

Flight Schedule Manager with AI Weather Assistant

- **Core:** Built an automated system integrating mobile OCR, a FastAPI server on AWS, the Google Calendar API, and an LLM.
- **Outcome:** Fully automated the process of parsing flight schedule photos, managing a calendar, and delivering real-time delay info and AI-powered weather briefings.
- **Github:** <https://github.com/taeyoungson/flight-schedule-parser>

AI-Powered Automated Stock Trader

- **Core:** Leveraged an LLM to perform natural language analysis on DART financial statements for automated stock selection.
- **Outcome:** Developed an end-to-end, AI-powered stock trading system that automates everything from stock selection to trade execution.
- **Github:** <https://github.com/taeyoungson/trader>

Honors & Awards

- CVPR Best Paper Finalist, 2022 (Awarded to Top 0.4%)
- Winner, Qualcomm Innovation Fellowship 2022
- Seoul MaaS Hackathon Excellence Award 2019 (우수상)

Media

- <https://news.mt.co.kr/mtview.php?no=2020091410243790785>
- <https://www.yeongnam.com/web/view.php?key=20220428010003756>
- <https://www.youtube.com/watch?v=LCV7OD3qUcQ>